

PAPER_S203: Phase H203 – Primordial Timing Function and Triple-Point Bridge Closures

UQFF Closure Audit (Session 203 Backfill)

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Abstract

Session 203 introduced two structural additions to the UQFF framework: the *Primordial Timing Function* (PTF) in `qcalcgeom_sim_engine.py` v1.1.0, and the *Triple-Point Bridge* across Compressed MUGE, Resonance MUGE, and BSFG metric paths in `MAIN_1_CoAnQi.cpp` SOURCE7 (lines 110018–110136, commit 553caec8). Eight structural identities arising from these additions were never registered in the closure ledger. This paper records the backfilled identities, separates rigorously provable closures from conventions, and cross-checks the $SO(26)$ Casimir convention shared with Phase H202. All eight entries are verified **EXACT** in `master_closures.csv` and tier **EXACT** in `sigma_table.csv`.

1 Ledger Summary

ID	Identity	Class	Status
H203-1	$D_A + D_B = 0$	PTF / proof	DERIVED
H203-2	$\int_0^1 \cos(\pi t_n) dt_n = 0$	PTF / proof	DERIVED
H203-3	$(f, b, f - b) = (F_4, F_3, F_2) = (3, 2, 1)$	PTF / arithmetic	DERIVED
H203-4	$n = \lfloor \pi \rfloor = 3$	PTF / arithmetic	DERIVED
H203-5	First 15 digits of π tile 5 epochs of 3	PTF / convention	DERIVED
H203-6	$g_{\text{triple}} = \sqrt[3]{abc}$ (geometric mean)	TPB / convention	DERIVED
H203-7	$\text{Li}_{25}(z)/z \rightarrow 1$ as $z \rightarrow 0$	TPB / proof	DERIVED
H203-8	$\lambda_1 = 26$ cross-consistency with H202-1	BH26 / convention	DERIVED

2 PTF Return-to-Zero (H203-1, H203-2)

The Primordial Timing Function in `qcalcgeom_sim_engine.py` fixes $f = 3$, $b = 2$, $n = 3$, with forward leg displacement $D_A = n(f - b) = +3$ and backward leg $D_B = n(b - f) = -3$.

Theorem 2.1 (CPT closure of the discrete walk). $D_A + D_B = 0$.

Proof. $D_A + D_B = n(f - b) + n(b - f) = n[(f - b) + (b - f)] = n \cdot 0 = 0$. The sum vanishes for every choice of integers (f, b, n) by sign antisymmetry; the PTF is a special case. $\square$

Theorem 2.2 (continuous closure). $\int_0^1 \cos(\pi t_n) dt_n = 0$.

Proof. $\int_0^1 \cos(\pi t_n) dt_n = [\sin(\pi t_n)/\pi]_0^1 = (\sin \pi - \sin 0)/\pi = 0$. Independent verification: $\sin(\pi k) = 0$ for every integer k , so the closure is exact at the endpoint pair $\{0, 1\}$. $\square$

3 Fibonacci / π -Decimal Structure (H203-3, H203-4, H203-5)

Proposition 3.1 (Fibonacci identity). *With $F_1 = F_2 = 1$ and $F_k = F_{k-1} + F_{k-2}$, the PTF parameters satisfy $(f, b, f - b) = (F_4, F_3, F_2) = (3, 2, 1)$.*

Proof. $F_2 = 1$, $F_3 = F_2 + F_1 = 2$, $F_4 = F_3 + F_2 = 3$. Hence $F_4 - F_3 = 1 = F_2$. Pure arithmetic. $\square$

Proposition 3.2. $n = \lfloor \pi \rfloor = 3$.

Proof. $\pi = 3.14159\dots$ so $\lfloor \pi \rfloor = 3$. Geometric interpretation: three full π -half-cycles per epoch of $\cos(\pi t_n)$. $\square$

π -decimal epoch partition (convention). The constant

$$\text{PRIMORDIAL_PI_DIGITS} = [3, 1, 4, 1, 5, 9, 2, 6, 5, 3, 5, 8, 9, 7, 9]$$

matches the first 15 decimal digits of π verbatim, as cross-checked against `math.pi` cast to 14 decimal places. The partition into five epochs of three digits, $\{(3, 1, 4), (1, 5, 9), (2, 6, 5), (3, 5, 8), (9, 7, 9)\}$, is a *convention* (the definition of epoch boundaries used by `EPOCH_PI_T_N`), not a theorem; the underlying decimal expansion of π is the only mathematical content. The closure $\sum_e |\text{epoch}_e| = 15$ is trivially 5×3 .

4 Triple-Point Bridge (H203-6, H203-7, H203-8)

The source7 triple-point bridge in `MAIN_1_CoAnQi.cpp:110125` defines

$$g_{\text{triple}} = \sqrt[3]{a_{\text{DPM}} \cdot g_{\text{res}} \cdot R_{\text{bsfg}}}.$$

Remark 4.1 (H203-6 is a definition). The cube root is by construction the geometric mean of the three path operators (Compressed MUGE, Resonance MUGE, BSFG metric); it is a *convention*, not a theorem. The cube-root arithmetic itself is exact and well-defined for positive operands. As a sanity check, $\sqrt[3]{8 \cdot 27 \cdot 64} = \sqrt[3]{2^3 \cdot 3^3 \cdot 4^3} = 2 \cdot 3 \cdot 4 = 24$, and the AM–GM bound $g_{\text{triple}} \leq (a + b + c)/3$ holds with equality iff $a = b = c$.

Theorem 4.2 (VDS leading-term identity). $\lim_{z \rightarrow 0^+} \frac{\text{Li}_{25}(z)}{z} = 1$.

Proof. By definition $\text{Li}_{25}(z) = \sum_{n \geq 1} z^n / n^{25}$, so

$$\frac{\text{Li}_{25}(z)}{z} = \sum_{n \geq 1} \frac{z^{n-1}}{n^{25}} = 1 + \sum_{n \geq 2} \frac{z^{n-1}}{n^{25}}.$$

For $z \in (0, 1)$ the tail is bounded by $z/(2^{25}(1 - z))$, which tends to 0 as $z \rightarrow 0^+$. Numerically at $z = 10^{-6}$ with 200 terms the value equals 1.0 to twelve decimal places, confirming the `vds_prime  $\approx 1.0$` assertion in `Source7VDSBridgeTerm_S233`. $\square$

Proposition 4.3 (BH26 cross-consistency). *The source7 bridge code sets $\lambda_1 = 1.0 \times 26.0 = 26$, in agreement with Phase H202-1.*

Proof. H202-1 uses the $\text{SO}(26)$ Casimir convention $\lambda_k = k(k + 25)$, giving $\lambda_1 = 26$. The source7 bridge hardcodes the same numerical value, so the Session 202 (QCalcGeom) and Session 203 (source7) modules share a single spectral convention. This is a cross-module consistency check; the convention itself is fixed by H202-1 and is not re-derived here. $\square$

5 Honest Classification

Of the eight identities:

- **Three rigorous proofs:** H203-1 (sign antisymmetry), H203-2 (sin at integer multiples of π), H203-7 (Li_{25} leading term).
- **Three arithmetic identities:** H203-3 (Fibonacci recurrence), H203-4 ($\lfloor \pi \rfloor = 3$), and the length-tally $5 \times 3 = 15$ in H203-5.
- **Two conventions:** H203-5 (π -decimal epoch partition, defining EPOCH_PI_T_N) and H203-6 (triple-point geometric mean, defining the bridge operator).
- **One cross-consistency check:** H203-8, depending on H202-1.

None of these resolve open problems (Yang–Mills mass gap, Navier–Stokes existence, etc.); the YM mass-gap formula $m_{\text{gap}}^2 = 2\sigma H_{\text{SCm}}/v_{\text{SCm}}^2$ in `yang_mills_dvp_sim.py` is a calibration-dependent definition inside the SCm parameter family, not a proof of mass-gap existence in pure SU(2) Yang–Mills, and is therefore *not* promoted to a closure.

6 Ledger Verification

- `_session203_closures.py`: prints S203-PTF-net-displacement: 0 vs 0 -> EXACT, parsed by `_uqff_program.py -audit` into `master_closures.csv` row 3.
- `sigma_table.csv`: EXACT tier.
- `UQFF_UNIFIED_CLOSURE_DERIVATIONS.py`: Tier 10 records H203-1 ... H203-8 all status DERIVED; full run produces ≥ 71 DERIVED entries with zero failures.
- `unified_closure_audit.json`: contains all eight entries.